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NMAP to discover live hosts using Nmap Scan (ARP, ICMP, TCP/UDP) on the tryhackme platform.

Aim :

To discover live hosts using Nmap scans such as ARP, ICMP, TCP and UDP scans on the tryhackme platform

Introduction :-

Nmap (Network Mapper) is a network scanning & host discovery tool. It is used to identify active systems in a network and gather information about the available hosts and ~~system~~ services. Nmap can use different techniques to discover whether a system is active. common host discovery techniques include ARP scan, ICMP scan & TCP/UDP ping scan.

Host discovery methods:

i) ARP scan:

ARP (address resolution protocol) is mainly used to discover host in the local network. An ARP scan sends ARP requests to the same local network.

ii) ICMP scan:

ICMP scan is commonly used to check whether a host is reachable. An ICMP scan sends ICMP request to the target host & checks for a response. If the host responds it can be identified as a live host.

iii) TCP / UDP scan:

TCP / UDP packets can be used for host discovery. We send suitable TCP or UDP packets to the target and analyses the response to determine whether the host is up

Procedure:

* Open the tryhackme Nmap environment
* Identify the target IP address or IP address range to be scanned.
* Start the Nmap host discovery process
* perform an ICMP scan to check the reachability of the target host
* perform TCP/UDP ping scans to discover hosts that responds to TCP/UDP packets.
* Observe the responses received from the target systems.
* Identify the hosts that are currently active or
* Record the discovered IP address and scan res
* Compare the results obtained using the different scanning methods.

Results:

Thus the live hosts in the target network we successfully discovered using Nmap ARP, ICMP & TCP/UDP host discovery scans.